

1(a)	any three from: 1 catch (a sample of grey bush) crickets (within habitat) and count total captured and mark and release them ; 2 detail ; e.g. <i>idea of</i> suitable marking so not harmful / too obvious / not removed e.g. <i>idea of</i> leaving for a few days (to give time for populations to mix before second sample taken) 3 catch second sample and count total number of crickets and number of marked crickets ; 4 reference to Lincoln index ;	3
1(b)	Table 1.1 completed correctly and $\Sigma(n / N)^2 = 0.21 / 0.22$ when answer rounded to 2 significant figures ; $D = 0.78 / 0.79$ when rounded to 2 significant figures ; $D = 0.783$; must be to 3 significant figures ;	3

Question	Answer	Marks
1(c)(i)	any two from: 1 wind, speed / direction ; 2 (atmospheric) humidity ; 3 rainfall / soil water content ; 4 soil type ; 5 soil pH ; 6 salinity (of soil) ; 7 mineral / ion, concentration (of soil) ; 8 (atmospheric) carbon dioxide concentration ;	2

Question	Answer	Marks												
1(c)(ii)	any seven from: 1 use quadrats ; 2 method of, random / systematic, placement of quadrats ; 3 use, same / stated, size (frame / point), quadrat ; 4 method of identifying (plant) species ; 5 count the number of individuals of each (plant) species ; 6 <i>idea of</i> taking care to note, low growing / small / easy to miss, species ; 7 <i>idea of</i> sampling, 10 or more / AW, separate points / quadrats (in the grey dune habitat) ; 8 (measure biodiversity of plant species) at regular intervals (over the 10-year period) and at the same time of the year ; 9 repeat (whole investigation over 10 years) at different, times of year / seasons ; 10 named hazard and risk and precaution ; <table border="1"> <thead> <tr> <th>hazard</th><th>risk</th><th>precaution</th></tr> </thead> <tbody> <tr> <td>plants / pollen</td><td>allergy / irritation / injury</td><td>gloves / eye protection / mask / PPE / protective clothing</td></tr> <tr> <td>animals</td><td>allergy / irritation / injury</td><td>gloves / eye protection / mask / PPE / protective clothing</td></tr> <tr> <td>grey dunes / ecosystem</td><td>getting lost / trip hazard / injury</td><td>wear suitable shoes / clothes / gloves / PPE stay, in a group / with expert / ranger</td></tr> </tbody> </table>	hazard	risk	precaution	plants / pollen	allergy / irritation / injury	gloves / eye protection / mask / PPE / protective clothing	animals	allergy / irritation / injury	gloves / eye protection / mask / PPE / protective clothing	grey dunes / ecosystem	getting lost / trip hazard / injury	wear suitable shoes / clothes / gloves / PPE stay, in a group / with expert / ranger	7
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